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Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.–3)

PHYSICAL PHARMACEUTICS-I

Subject Code : BP-302T

M.Code : 75106

Date of Examination : 11-12-2023

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. **Define briefly :**

   a. What is meant by contact angle and how is it related to wetting of solids?

   b. What are polar and semi-polar solvents? Give two examples each.

   c. What is meant by salting out?

   d. What is meant by internal pressure?

   e. What is spreading coefficient?

   f. Define 'freezing point depression'.

   g. What is self-association?

   h. What is an isotonic solution?

   i. What are eutectic mixtures?

   j. What are clathrates? Give two examples.

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SECTION-B

2. Explain the phase diagram for water at moderate pressures and discuss the occurrence of "triple point".

3. Discuss ideal solutions and explain complete and partial miscibility.

4. Classify complexes with examples. Discuss one method to determine the stoichiometry of complexes.

SECTION-C

5. Write a note on influence of buffer capacity and pH on tissues.

6. Discuss the solubility method for determining complex formation.

7. Write a note on ampholytes with examples.

8. Explain hydrophobic interaction and its significance.

9. Write a note on distribution coefficient.

10. What is dissociation constant? Write a note on it and explain its significance.

11. What are liquid crystals and what are their applications?

12. Summarize the application of diffusion in drug absorption.

13. Differentiate between amorphous and crystalline solids with respect to solubility and dissolution of drug molecules.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

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